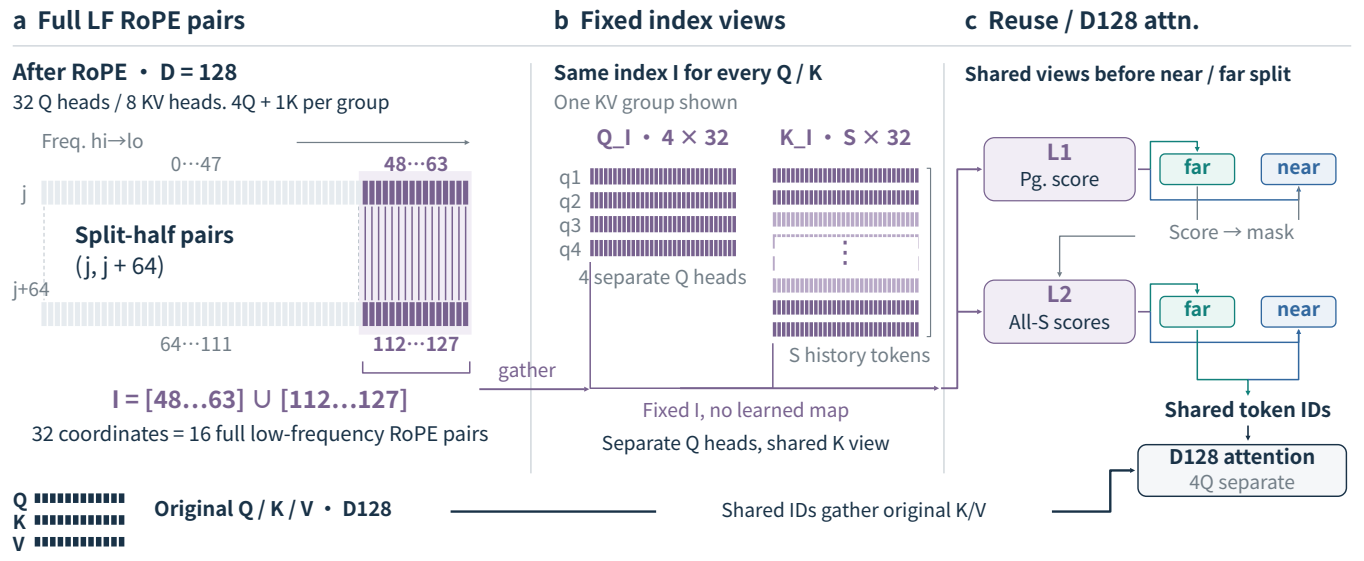

a Full LF RoPE pairs
After RoPE • D = 128
32 Q heads / 8 KV heads. 4Q + 1K per group
Freq. hi→lo
0···47
48···63
j
Split-half pairs
(j, j + 64)
j+64
64···111
112···127
I = [48…63] ∪ [112…127]
32 coordinates = 16 full low-frequency RoPE pairs
b Fixed index views
Same index I for every Q / K
One KV group shown
Q_I • 4 × 32
K_I • S × 32
q1
q2
q3
q4
4 separate Q heads
S history tokens
gather
Fixed I, no learned map
Separate Q heads, shared K view
c Reuse / D128 attn.
Shared views before near / far split
L1
Pg. score
far
near
Score → mask
L2
All-S scores
far
near
Shared token IDs
D128 attention
4Q separate
Q
K
V
Original Q / K / V • D128
Shared IDs gather original K/V